Number Spiral

Time limit: 1.00 s
Memory limit: 512 MB

CSES 1071 | Introductory Problems | https://cses.fi/problemset/task/1071

A number spiral is an infinite grid whose upper-left square has number 1. Here are the first five layers of the spiral: Your task is to find out the number in row y and column x.

1	2	9	10	25
4	3	8	11	24
5	6	7	12	23
16	15	14	13	22
17	18	19	20	21

Input

The first input line contains an integer t: the number of tests.

After this, there are t lines, each containing integers y and x.

Output

For each test, print the number in row y and column x.

Constraints

- $1 \le t \le 10^5$
- $1 \le y,x \le 10^9$

Example

Input

```
3
2 3
1 1
4 2
```

Output

```
8
1
15
```